Neural Stiff-Chemistry Surrogates in DeepFlame: Implementation Pitfalls, Coupled-Solver Safeguards, and Target Semantics for H_2 and C_2H_4

Project draft generated from experiment record

April 24, 2026

Abstract

This draft manuscript summarizes the current state of an EFNO-oriented stiff-chemistry implementation and deployment program for H_2 and C_2H_4 inside DeepFlame. The strongest present story is not a final claim of solved CFD-coupled chemistry replacement, but an evidence chain showing: (1) EFNO-style stiff-chemistry surrogates are highly sensitive to implementation details, including a consequential transformed-state decode bug; (2) offline accuracy and rollout metrics do not directly predict coupled DeepFlame usefulness; (3) H_2 deployment requires thermodynamic diagnostics and guarded hybrid policies to remain solver-usable; and (4) for the more difficult C_2H_4 case, data construction and label semantics appear to dominate the remaining error budget. The manuscript is therefore organized as a reviewable partial-results paper with explicit negative results, deployment failures, and reproducible artifacts.

1 Introduction

Stiff chemistry surrogates are attractive for CFD because conventional chemistry integration remains expensive, but the deployment-facing evidence base for neural surrogates is still thin [2, 1, 4]. This project targets a faithful EFNO-style implementation study for H_2 and C_2H_4 inside DeepFlame, with emphasis on what survives coupled deployment rather than only what looks good offline.

2 Background and related work

The manuscript is grounded in the EFNO paper by Weng et al. [3] and a local-paper crosswalk covering data generation, augmentation, and coupled-validation issues from related combustion-chemistry learning papers [2, 5, 1, 4]. Across these papers, a recurring theme is that model architecture alone is not decisive: data semantics, postprocessing contracts, augmentation strategy, and a posteriori validation all materially affect conclusions.

3 Reproducible workflow and software stack

The implementation workflow combines PDF extraction and structured notes, DFODE-kit extensions for EFNO-style training and evaluation, and DeepFlame export bridges for MLP and FNO-like surrogates. The software path is deliberately aligned with the emerging DFODE-kit/DeepFlame ecosystem [1, 2]: data are prepared from canonical or case-aligned thermochemical states, models

are trained and evaluated offline with explicit rollout and physical-consistency metrics, and promising checkpoints are exported into DeepFlame for coupled testing. All substantive experiments are documented in the project docs site together with scripts and artifact paths.

4 Methods and evaluation protocol

The study is organized as a staged implementation-and-deployment workflow rather than a single train/test exercise. The offline H_2 benchmark uses autoignition trajectories generated from a seven-species hydrogen mechanism, with holdout splits defined by initial-condition trajectories rather than random sample shuffling so that rollout claims reflect unseen trajectory behavior rather than interpolation within the same paths. Models are compared with fixed seeds and evaluated on one-step thermochemical error, long-horizon rollout error, and physical-consistency metrics including element-mass and mass-sum deviations. This choice follows directly from the EFNO paper's emphasis on autoregressive prediction under unseen conditions [3], but the present implementation also makes the evaluation contract explicit because several apparently strong branches change rank once deployment-facing postprocessing is enforced.

For DeepFlame-coupled H_2, the target case is the Burke nine-species configuration rather than the earlier seven-species offline mechanism. That mechanism/species-count alignment became a hard requirement once export began. Coupled evaluations are therefore performed on Burke-aligned exported checkpoints and interpreted under the same species postprocessing contract used by the runtime bridge. In practice, this means distinguishing the project-side closure reconstruction from the DeepFlame-facing "preserve-last" rule, where the last species is preserved from the current state and the predicted main species are renormalized. This contract choice materially compresses some offline ranking gaps and is therefore part of the method, not an implementation footnote.

For C_2H_4, the baseline is a stock-style DeepFlame run in the trustworthy GPU regime (‘mpirun -np 8‘, ‘coresPerNode 8‘) carried to 5×10^{-6}. Learned-model experiments are only interpreted after this runtime baseline and after the exported FNO bridge was repaired to batch active-set inference, because earlier reduced-rank GPU crashes and late out-of-memory failures showed that runtime pathologies could otherwise be mistaken for model failures. The main C_2H_4 learned datasets then span three increasingly deployment-relevant families: crude case-pair labels extracted from CFD states, filtered/mixed case-pair subsets such as $dp100$, and canonical/augmented data motivated by nearby literature on reactive-region coverage [2, 5, 1].

Across both fuels, the manuscript prioritizes deployment-facing metrics over raw fit alone. Offline results are summarized by one-step MAE and rollout MAE, but coupled H_2 decisions are further driven by HP-reconstruction failure fractions, learned-vs-fallback active-cell fractions, and threshold/guard operating windows. Coupled C_2H_4 decisions are driven by survival to the stock baseline horizon together with matched-mesh comparisons of $\dot{Q}$, temperature, pressure, and key species. This method deliberately treats negative results—for example, preserve-last rebaselining, pure chemistry-proxy early failure, and random partial relabeling regression—as evidence about evaluation contracts and target semantics rather than as dead ends.

5 Offline H_2 reproduction and implementation diagnosis

The H_2 program established seeded offline baselines, rollout-aware EFNO-style training, and a key implementation diagnosis: a transformed-state species decode bug materially distorted rollout behavior. After correcting the update rule to act in the proper transformed state space, self-rollout behavior changed qualitatively and the corrected branch became a credible offline reference for

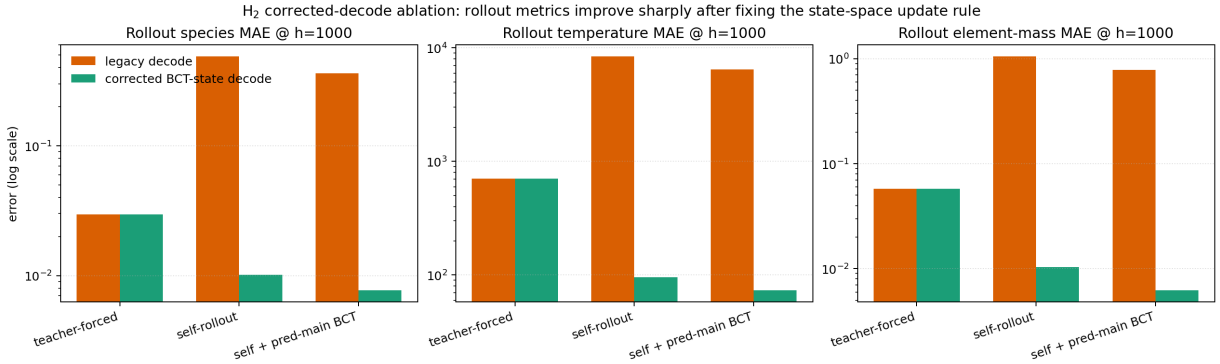

Figure 1: H_2 legacy-vs-corrected decode ablation on rollout metrics at horizon 1000. Fixing the transformed-state update rule changes the empirical conclusion most strongly for the self-rollout branches, supporting the claim that a large part of the earlier instability was implementation-induced rather than intrinsic to self-feeding.

Table 1: Selected H_2 coupled-deployment operating points for the corrected branch in DeepFlame.

Mode	Learned frac. at 2×10^{-5}	Cum. fallback frac. at 2×10^{-5}	First time > 95% fallback
FT550	0.000	0.516	1.4e-05
FT600	0.081	0.373	–
FT650	0.294	0.361	–
FT650+risk@0.5	0.241	0.381	–

deployment-facing work. Figure 1 shows that the correction was not a small numerical cleanup: it sharply reduced long-horizon rollout error, especially for the self-rollout branches.

6 DeepFlame-coupled H_2: from failure to guarded deployment

The H_2 coupled study showed that export correctness and offline metrics are not enough. Coupled failures appeared first as HP reconstruction breakdowns, which led to hybrid fallback policies, frozen-temperature gating, and a logistic risk guard. Figure 2 shows how threshold and guard choices reshape the coupled operating regime over time. Table 1 summarizes representative operating points for the corrected branch. In the present evidence base, 650 K gives the strongest long-horizon single-threshold retention, while the logistic risk guard at threshold 0.5 provides a more safety-oriented mode with lower HP-failure exposure at some cost in learned participation.

The H_2 evidence supports a reviewable claim that deployment-facing chemistry surrogates require contract alignment and explicit thermodynamic safeguards, not just better offline loss values.

7 C_2H_4 baseline establishment and runtime bridge diagnosis

For C_2H_4, the program first established a trustworthy stock DeepFlame baseline, diagnosed reduced-rank GPU pitfalls, and repaired the FNO runtime bridge by batching inference. These steps were necessary before data-quality conclusions could be trusted: without them, runtime pathologies could masquerade as model deficiencies. The emphasis on a posteriori validation follows the practice advocated in recent combustion-surrogate studies, which consistently show that apparently

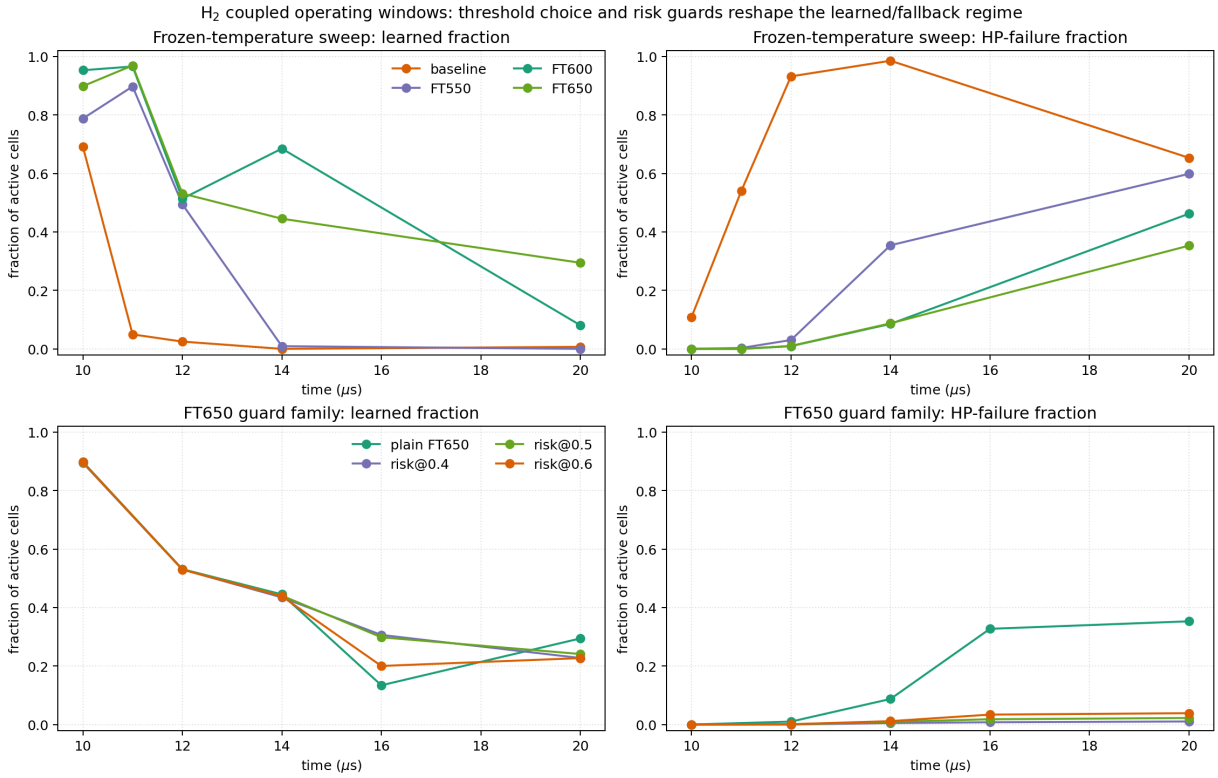

Figure 2: H_2 coupled operating windows from the corrected branch. Top row: frozen-temperature sweep showing learned-cell retention and HP-failure fractions over time. Bottom row: FT650 risk-guard family. The figure makes two manuscript-level points explicit: raising the activation threshold delays collapse materially, and the logistic risk guard trades some learned participation for much lower HP-failure incidence.

strong offline models can still fail once coupled back into CFD [2, 5, 4].

8 C_2H_4 data semantics: what the current best model still gets wrong

The strongest current C_2H_4 data result is a calibrated mixed dataset that combines the solver-aligned $dp100$ case-pair backbone with a moderate canonical augmentation component. This mixed model survives to 5×10^{-6} and is substantially better than earlier baselines, but the best-model-vs-stock anatomy shows a structured residual failure: cooler active cells are over-driven, the hot bulk exhibits widespread source-term sign errors, and several late intermediates are still heavily suppressed. The mixed-data direction itself is motivated by nearby literature showing the value of canonical flame manifolds, interpolation-based densification, and physics-aware perturbation for building thermochemical training sets that cover reactive regions more effectively [2, 5, 1].

The next diagnostic step showed that the target-semantics hypothesis is real in the data itself: one-step chemistry-proxy relabeling of case-pair current states produces materially different targets, especially on key intermediates. Table 3 summarizes the most distorted active-region species in the current best mixed model, while Table 4 shows that the original CFD next-state labels differ materially from chemistry-proxy relabels on a sampled subset. However, deployment tests also

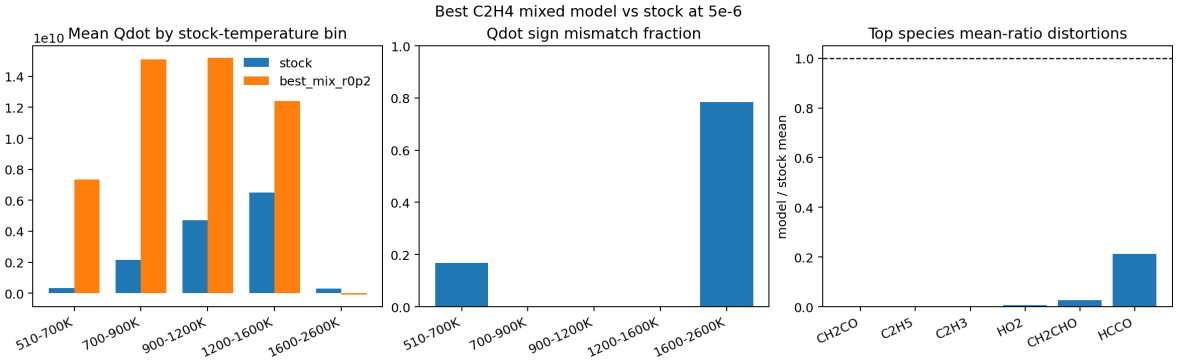

Figure 3: Matched-mesh comparison between the current best mixed C_2H_4 model and the stock DeepFlame baseline at 5×10^{-6}. The figure highlights temperature-binned source-term mismatch and the strongest mean-ratio species distortions.

Table 2: C_2H_4 deployment-facing summary at 5×10^{-6} for the stock baseline, the current best mixed model, and a naive partial chemistry-proxy relabeling ablation.

Model	Mean $\dot{Q}$	$\dot{Q}$/stock	$p_{\max}$ (Pa)	$T_{\min}$ (K)	mean $\lvert\Delta T\rvert$ (K)
Stock DeepFlame	1.62e+07	1.000	1.02e+05	499.249	0.828
Best mixed (*dp*100+ *canon*@0.2)	2.51e+07	1.551	1.03e+05	499.159	0.927
Partial chem-proxy (10%)	-9.88e+08	-60.952	1.03e+05	492.524	9.153

showed that semantics-only fixes are not enough in isolation. A tiny pure chemistry-proxy model fails catastrophically by 4×10^{-7}, while naive partial replacement of 10% of labels preserves runtime survival but badly degrades solver-facing quality.

Table 2 makes the current C_2H_4 claim boundary explicit. The best mixed model is reviewably better than prior baselines, but naive partial chemistry-proxy relabeling is not yet a deployment improvement. Tables 3 and 4 sharpen the interpretation: the remaining gap is tied both to suppressed intermediate chemistry in deployment and to a measurable mismatch between full-CFD and chemistry-only target semantics. This supports the more specific forward hypothesis that semantics must be improved without losing the solver-relevant manifold.

9 Discussion

The present evidence supports several broader lessons. First, transformed-state implementation details can dominate empirical conclusions, which reinforces the need to treat reproduction as implementation diagnosis rather than a narrow hyperparameter chase [3]. Second, offline and deployment-facing evaluations can disagree substantially because of postprocessing and thermodynamic contract issues, in line with the broader combustion-surrogate literature's emphasis on a posteriori validation [2, 4]. Third, for the more difficult C_2H_4 case, the dominant remaining bottleneck appears to be target construction rather than another round of small architectural or mixing sweeps.

Table 3: Largest active-region species distortions for the current best mixed C_2H_4 model relative to stock at 5×10^{-6}.

Species	Stock mean	Model mean	Model/stock
CH2CO	2.12e-04	2.95e-17	1.39e-13
C2H5	1.50e-05	2.22e-16	1.47e-11
C2H3	5.81e-05	3.45e-10	5.95e-06
HO2	5.46e-05	3.64e-07	0.007
CH2CHO	6.63e-06	1.77e-07	0.027
HCCO	6.01e-06	1.28e-06	0.212

Table 4: Largest mean-ratio shifts between original CFD next-state labels and chemistry-proxy relabels on a 5k-row C_2H_4 subset.

Species	Original next mean	Chemistry-proxy next mean	Chem/original
HCCO	3.82e-07	1.34e-05	35.146
C2H5	5.13e-06	2.12e-07	0.041
CH2CHO	2.54e-06	1.08e-05	4.269
CH2OH	1.38e-06	5.13e-06	3.713
C2H3	3.90e-05	1.03e-04	2.651
CH2CO	6.24e-05	1.64e-04	2.621

10 Conclusion

The current project state is strong enough for a reviewable partial-results paper. For H_2, corrected implementation plus guards yields a credible deployment story. For C_2H_4, the strongest current result is a calibrated mixed-data baseline together with a clear diagnosis that better target semantics are needed, but must be introduced without abandoning the solver-relevant state manifold. The central take-home message is that offline surrogate success is insufficient without deployment-aware validation and explicit thermodynamic diagnostics.

Immediate remaining tasks before submission

1. Add citations from the extracted EFNO notes and local related papers.
2. Package at least one H_2 offline corrected-decode figure and one H_2 coupled-timeline figure.
3. Expand the manuscript body from a structured skeleton into a fully referenced narrative.
4. Replace naive semantics-only C_2H_4 relabeling with a more selective semantics-improvement experiment, or scope that step explicitly as future work.

References

[1] Han Li, Ke Xiao, Yangchen Xu, Haoze Zhang, Zhenyi Chen, Runze Mao, and Zhi X. Chen. DFODE-Kit: Deep learning package for solving flame chemical kinetics with high-dimensional stiff ordinary differential equations. *Computer Physics Communications*, 321:110013, 2026.

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